

Dr. Ama Koranteng

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<https://amakora0.github.io>

EDUCATION	Johns Hopkins University PhD , Computer Science <i>Advisor: Michael Dinitz</i> <i>Thesis: Approximation Algorithms for New Problems in Network Design</i>	June 2025
	Johns Hopkins University MSE , Computer Science	Dec 2022
	Massachusetts Institute of Technology BS , Mathematics	Feb 2020
WORK EXPERIENCE	Boeing — Applied Mathematician	Nov 2025 – present
	Boeing — Applied Mathematics Intern <i>Supervisor: Dr. Richard Joel Thompson</i> - Supported development of quantum heterogeneous computing resource estimator tool - Wrote software to take in Python-implemented algorithms and output directed acyclic graph (DAG) representations of the algorithms - Developed mathematical mixed-integer linear program formulation for algorithm workflow optimization, based on the DAG representations	Jun – Sep 2024
	Northrop Grumman — Intern <i>Supervisor: Dr. Jonathan Las Fargeas</i> - Studied and modified existing distributed algorithms for collaborative drones - Implemented algorithms in C# using behavior trees in Unity game engine	Jun – Aug 2017
	Northrop Grumman — Intern <i>Supervisor: Dr. Jonathan Las Fargeas</i> Created MATLAB models and Java programs to model complex offensive and defensive aircraft combat maneuvers	Jun – Aug 2016
	Johns Hopkins University — Doctoral Researcher - Developed algorithms for graph fault-tolerance and minimization optimization problems, resulting in several publications - Communicated research results to diverse audiences at several international and domestic conferences and seminars - Taught, mentored, and tutored high school, undergraduate, and junior PhD students in algorithms research, Python programming, and career development	Sep 2020 – present
RESEARCH EXPERIENCE	MIT Theory of Computation Group — Undergraduate Researcher <i>Advisor: Dr. Jayson Lynch</i> - Read and synthesized papers on gadget abstractions used to prove computational complexity of various games - Rigorously defined and reconciled new and existing frameworks and gadget abstractions - Solved open problems related to these papers, including designing algorithms and proving hardness results	Jun 2018 – May 2019
PUBLICATIONS		

with Michael Dinitz and Yasamin Nazari

Approximation Algorithms for Optimal Hopsets

International Colloquium on Automata, Languages, and Programming (ICALP), 2025

with Greg Bodwin, Michael Dinitz, and Lily Wang

Light Edge Fault Tolerant Graph Spanners

International Colloquium on Automata, Languages, and Programming (ICALP), 2025

with Michael Dinitz, Guy Kortsarz, and Zeev Nutov

Improved Approximations for Relative Survivable Network Design

Workshop on Approximation and Online Algorithms (WAOA), 2023

with Michael Dinitz and Guy Kortsarz

Relative Survivable Network Design

International Conference on Approximation Algorithms for Combinatorial Optimization Problems (APPROX), 2022

AWARDS

NSF Graduate Research Fellowship (acceptance rate $\sim 15\%$)	2022
GEM Fellowship (declined) (acceptance rate $\sim 10\%$)	2022
Graduate Fellowship for STEM Diversity (declined)	2022
Wu and Zhang Endowed Graduate Student Fellowship	2020
Google Computer Science Research Mentorship Program	2019
Bloomberg Grace Hopper Travel Grant	2016
Generation Google Scholarship (acceptance rate $\leq 1\%$)	2014
NCWIT Aspirations in Computing Award	2014

SERVICE

Mentor , Institute for African-American Mentoring in CS (IAAMCS)	Oct 2024 – present
• Provide guidance to computer science undergraduate students on applying to PhD programs, conducting research, and finding work/life balance	
Tutor , Intro to Algorithms	Oct 2024 – Feb 2025
• Taught 1-on-1 interactive lessons on introductory concepts in algorithms and discrete mathematics	
Advisor and Mentor , JHU Whiting Internships in Science and Engineering	Jun 2023 – May 2024
• Advised and mentored a Baltimore city high school student • Introduced student to basic graph theory and algorithms • Guided student through a programming project in which they implemented graph algorithms in Python • Mentored several other students in the WISE program	
Conference Reviewer	
• ICALP 2025, 2023 • STACS 2025 • ESA 2024, 2023 • SIROCCO 2023	
PhD Representative , JHU CS Diversity and Inclusion Committee	Jan – Dec 2022
• Worked with department administrators to help start and facilitate the JHU CS PhD Mentor Hour series, where senior PhD students give advice to junior students on a variety of topics in a casual, discussion-based setting	

Co-Organizer, JHU CS Theory Seminar Jan – May 2022

- Networked and liaised with CS theory researchers throughout the country, inviting them to give talks at JHU

Teaching Assistant, Algorithmic Game Theory (JHU) Jan – May 2022

- Held on-demand office hours to assist students with concepts and assignments
- Created rubrics and graded assignments

Peer Mentor, MIT Student Support Services 2017 – 2019

- Provided guidance as well as emotional and logistic support for undergraduate students through their leaves of absence

Assistant Educator, Sarah’s Science Feb – Aug 2016

- Introduced kindergarten students to basic science concepts through hands-on activities

Organizer and Educator, MIT Educational Studies Program 2014 – 2016

- Co-organized the MIT Summer HSSP program, a summer extracurricular education program for over 100 middle and high school students in the Boston area
- Co-organized the MIT Cascade program, a free after-school program for over low-income Boston-area high school students
- Interviewed and hired teachers, taught courses, and advertised for both programs

TECHNICAL TALKS

Light Edge Fault Tolerant Spanners
ICALP 2025, Aarhus, Denmark July 2025

Approximation Algorithms for Optimal Hopsets
ICALP 2025, Aarhus, Denmark July 2025

Relative Survivable Network Design
JHU CS Theory Seminar, Baltimore, MD, USA Nov 2023
Capital Area Theory Seminar (CATS), College Park, MD, USA Nov 2023

Improved Approximations for Relative Survivable Network Design
WAOA 2023, Amsterdam, The Netherlands Sept 2023

Relative Survivable Network Design
APPROX 2022, Online Sept 2022

SKILLS

Programming: Python, Java, $\LaTeX$
Technical: combinatorial optimization, network/routing optimization, theoretical computer science, discrete mathematics, algorithms and data structures, graph theory, graph algorithms, mixed integer programming, mathematical programming, Git, research
Other: teaching, tutoring, technical writing, collaboration, mentorship, communicating technical content to diverse audiences

AFFILIATIONS

Institute for African American Mentoring in Computing Sciences (IAAMCS), Society for Industrial and Applied Mathematics (SIAM), National Center for Women and Information Technology (NCWIT), Rewriting the Code (RTC)